HEALTHCARE SYSTEMS PROCESS IMPROVEMENT CONFERENCE 2026

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LEADING HEALTHCARE IMPROVEMENT

Smarter Early-Warning for Newborns: Reducing Delays and Alarm Fatigue in the NICU

(Designing AI Systems That Clinicians Can Actually Use)

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Key Learning Objectives

Session Overview:

- Why smarter early-warning is urgently needed in the NICU
- What we built using the monitor data hospitals already collect
- How it performed and what it means for safety and workload
- We focus not on building a better algorithm, but on designing a better warning system.

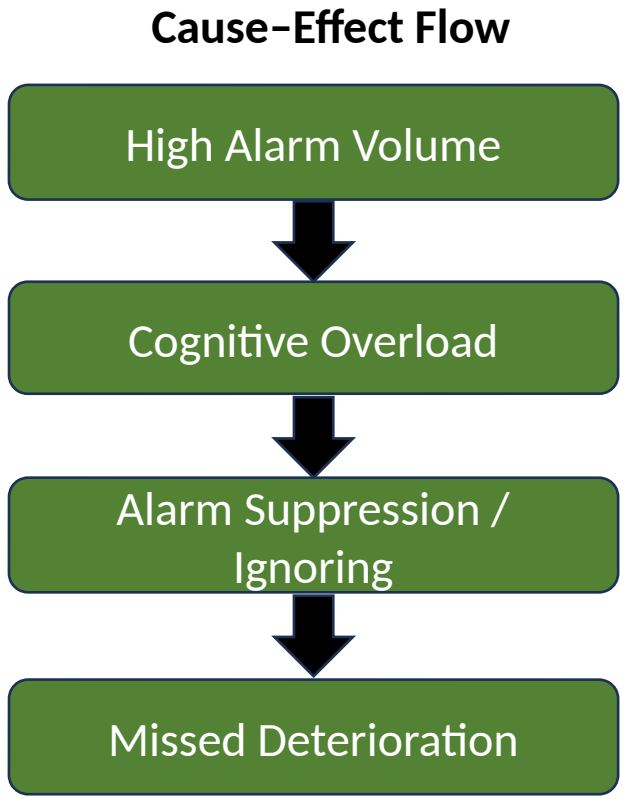
After this session, attendees will be able to:

- Explain how smarter monitoring improves newborn safety while reducing alarm fatigue
- Describe how to set and maintain alarm thresholds that balance early detection with workload
- Apply this framework to other settings where early warning and workflow integration matter



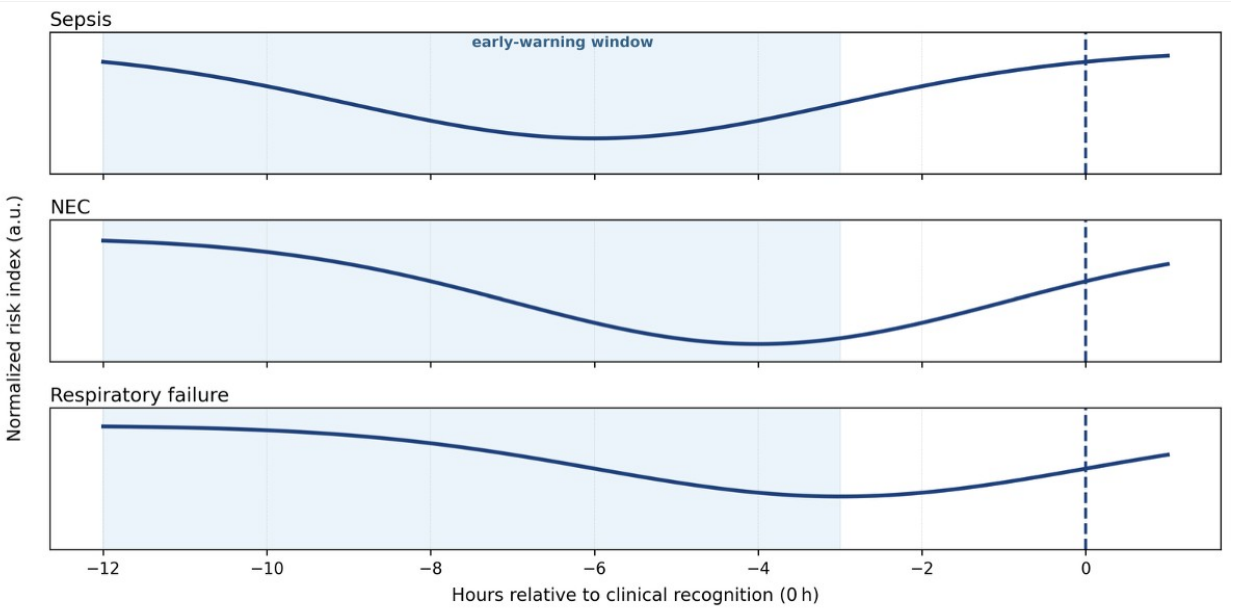
It isn’t Detection, It’s Deployment

- Alarm rates exceed human processing capacity
- Clinicians learn to ignore alarms to survive the shift
- Adding “better AI” often makes the problem worse
- **Result:** missed deterioration despite more technology



Why This Project?

- Newborns can deteriorate quickly, and late infection recognition leads to serious complications
- Existing monitors produce many non-specific alarms, driving alarm fatigue
- This project uses existing NICU monitor data to detect deterioration earlier with fewer alarms
- The goal is to give clinicians more lead time and clearer, more trustworthy alerts



Process Improvement Lens

Aim: Earlier, more trustworthy warnings at the same alarm burden
Alarm burden defines feasibility, lead time defines value

Outcome Measures:

Optimization Objective

- Lead time to recognition (Primary Obj.)
- Share of events with early warning*

Process Measures:

Hard Constraint

- False alarms per patient-day (FAPD) (Constraint not optimized)
- Time from alert to reassessment*

Balancing Measures:

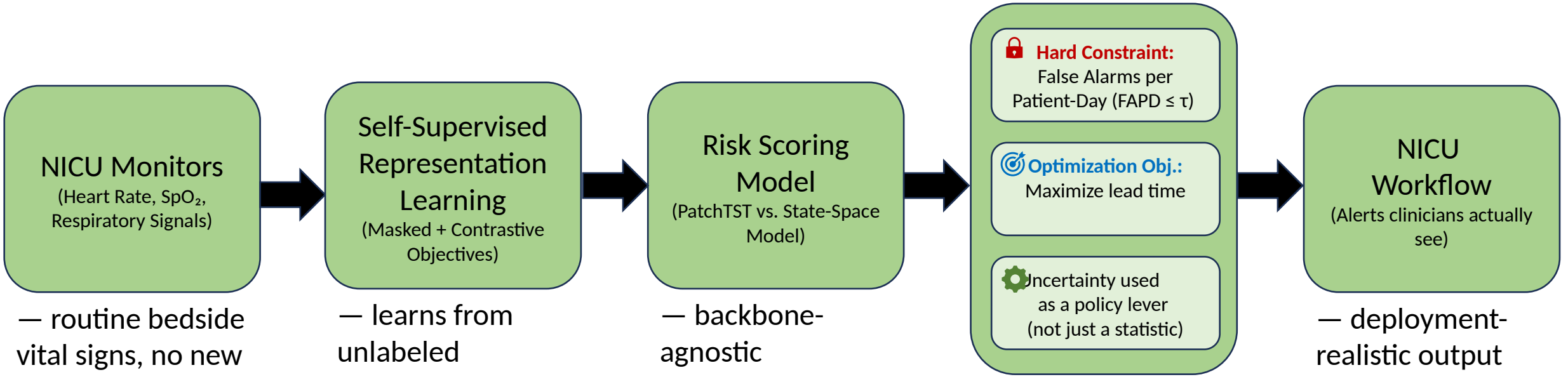
(Protected by system constraints)
(Maintained through constraint enforcement)

- Nurse alarm fatigue & trust*
- Workflow disruption

*(Secondary/derived outcome)

Data & Approach

Requirement first, model second
Alarm burden defines feasibility, lead time defines value



— routine bedside vital signs, no new sensors

— learns from unlabeled physiologic data

— backbone-agnostic comparison

— deployment-realistic output

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Performance Under Equal Alarm Burden

Same workload → different clinical value

Models are compared at the same false-alarm burden, not at arbitrary thresholds.

Hard Constraints (Fixed Across Models)

- False alarms per patient-day (FAPD) ≤ τ
- Refractory alerting applied identically
- Same patients, same windows, same evaluation protocol

Non-negotiable system constraint

Primary Obj. (Optimized)

Earlier warning under fixed alarm burden

Secondary/emergent properties: (not optimized)

- Calibration stability
- Threshold Robustness

Observed Performance at Matched Alarm Burden

PatchTST (SSL Transformer)

- ☑ Meets FAPD constraint
- ⊕ ~20 minutes median lead time

Mamba (State-Space)

- ☑ Meets FAPD constraint
- ⚠ Baseline/shorter median lead time
- ~20 minutes median lead time

Under identical workload constraints, architectural choice materially changes the feasible operating region of the alerting system.

Key Results

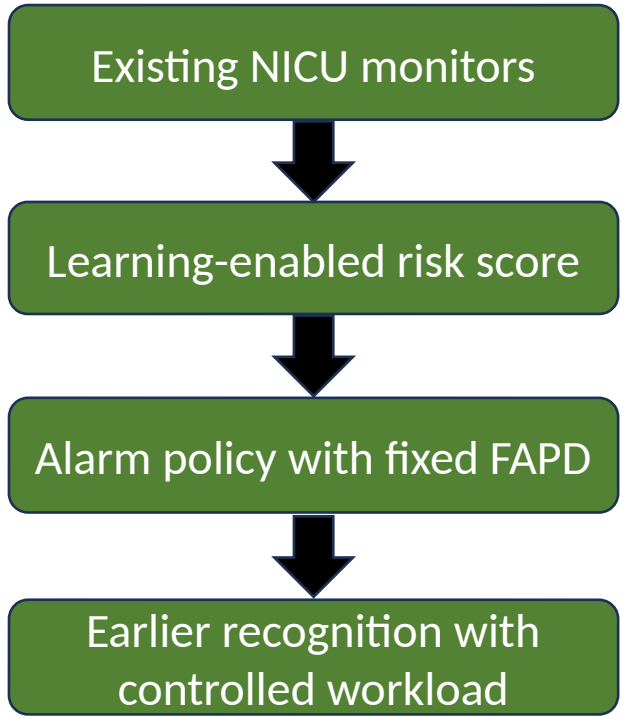
- Achieved earlier median warning time under fixed alarm-burden constraints
- Achieved earlier warnings without increasing alerts per patient-day, preserving workload feasibility
- Stable performance observed across evaluation runs and operating points
- Alerts showed which signals changed and when, improving transparency
- **Goal:** more useful warning time, not more alarms

Outcome Dimension	PatchTST (SSL Transformer)	Mamba (SSM)	Why This Matters
Alarm burden (FAPD)	≤ τ (met)	≤ τ (met)	Ensures fair comparison
Median lead time	+~20 min earlier	Baseline	More time to act
% events ≥ 30 min early	Higher	Lower	Practical early warning
Calibration stability	Stable across thresholds	Threshold-sensitive	Predictable behavior
Threshold robustness	Low sensitivity	High sensitivity	Safer configuration
Alert timing variance	Smooth	Erratic	Trust & interpretability
Operational risk	Lower	Higher	Deployment safety

Practical Implications & Next Steps

- Alarm burden can be treated as a design constraint, not a tuning artifact
- Model choice changes the feasible operating region under fixed workload
- Earlier warning does not require more alarms
- System stability matters as much as raw predictive power

From Research to Deployment



Thank you!